higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 3

(12020353)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1: GENERAL

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 … is a mixture of cement, aggregate (sand and stones) and water.

A Grout
B Lime
C Concrete
D Mortar

1.2 A … wall has TWO LEAVES parallel to each other.

A soldier bond
B broken bond
C single leaf
D cavity

1.3 … is a structure that carries brickwork on soil or earth.

A A foundation
B A beam
C A column
D Scaffolding

1.4 … is a piece of metal that connects leaves of masonry to each other or to other materials.

A A lintel
B A ring beam
C A wall tie
D Corrugated iron

1.5 A … is an opening placed in the mortar joints of facing materials at the level of flashing, to permit moisture to escape.

A weep hole
B trap door
C ventilation door
D cavity wall

1.6 To make a concave joint, use a … jointer.

A round
B flat
C V-
D sharp

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1.7 A soffit is the underside or bottom of a …

A foundation.
B column.
C arch.
D profile.

1.8 Leave the supports for the bed-joint reinforced lintels for at least … days after they have been completed.

A 2
B 100
C 50
D 7

1.9 The … is the distance between the springing line and the highest point of the arch.

A intrados
B key brick
C rise
D span

1.10 The … is the small hollow underneath a brick.

A core
B cavity
C opening
D frog

(10 × 1) **[10]**

QUESTION 2: CAVITY WALL CONSTRUCTION

2.1 Draw, to a scale of 1:10, a vertical section through the bottom eight (8) courses of a 270-mm cavity wall construction. Show the following details:

- Concrete strip foundation, 600 mm × 200 mm
- Concrete slab, 75 mm
- Plaster, 15 mm (internally only)
- Screed, 20 mm
- Hardcore
- Ground level
- Wall tie
- DPC at floor level height
- Cavity filled with concrete up to DPC

NOTE: Use a standard brick of 220 mm × 110 mm × 75 mm. (20)

2.2 What are the minimum and the maximum width of a cavity wall? (2)

2.3 What is the purpose of brick force? (2)

2.4 State FOUR disadvantages of a cavity wall. (4)

2.5 In what direction should the wall ties be inserted? (2)
[30]

QUESTION 3: BASIC MASONRY WALL CONSTRUCTION WITH HOLLOW UNITS

3.1 Briefly explain the terms *jointing* and *pointing.* (2 × 2) (4)

3.2 Name FIVE different types of jointing methods. (5)

3.3 List FOUR types of foundation. (4)

3.4 Identify FIVE different types of hollow masonry units and indicate where the units will be used. (5 × 2) (10)

3.5 What is the difference between a *gauge rod* and a *corner profile*? (2 × 2) (4)

3.6 Write down an explanation of the following terms:

3.6.1 Return wall

3.6.2 Cross wall

3.6.3 Conduit

(3 × 1) (3)
[30]

QUESTION 4: ADVANCED MASONRY WALL CONSTRUCTION

4.1 Briefly explain how to rake out and rule bed joints. (3)

4.2 Draw, to a scale of 1:10, the front view of a semi-circular gauged arch and clearly show the following:

- Span, 1 000 mm
- Key brick
- Voussoirs
- Extrados
- Intrados
- Springing point
- Springing line
- Depth of arch, 220 mm

NOTE: Use a standard brick of 220 mm × 110 mm × 75 mm. (20)

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4.3 Name TWO types of brick force used to reinforce masonry walls and walls across openings. (2)

4.4 What are the specified depths of roof anchors for lightweight and heavyweight roofs? (2)

4.5 Arches must be constructed according to the regulations/guidelines structured in the National Building Standards (SANS 10400-K).

List THREE of these regulations/guidelines. (3)
[30]

TOTAL: 100

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